MRI Foreground/Background Detection

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Abstract

This note outlines how we derive foreground and background masks for MRI volumes in RadQy. Each section documents one detection method, its assumptions, and the clean-up steps needed so FG/BG pixels are reproducible across datasets.

Contents

1 Introduction

This document lists the foreground/background (FG/BG) segmentation methods in `backend/seg/mri`. Each section mirrors a Python module and notes its inputs, thresholds, morphology, and fallbacks so downstream IQMs get consistent masks.

2 Image Processing Segmenters

Classical pipelines based on intensity statistics and morphology.

2.1 OtsuHull (`OtsuHull.py`)

Legacy variant mirroring the original hybrid Otsu pipeline with more defensive fallbacks [1]. This subsection is listed first because it matches the earliest implementation shipped with RadQy.

- **Preprocess**: Cast to uint16. Histogram-equalize (uses `scikit-image` if available; otherwise internal equalization on $[0, 1]$).
- **Dual thresholds**: Otsu on raw and equalized images to get masks and FG fractions w_1, w_2.
- **Re-threshold**: Weighted sum of raw and equalized images, followed by Otsu.
- **Hull cleanup**: Convex hull if available; otherwise hole-filling. Any exception yields FG all-ones, BG all-zeros.
- **Output**: `segment(img)` returns FG; `make_masks(img)` returns (fg, bg) in $\{0, 1\}$.

3 Deep Learning Segmenters

Learned models for FG/BG inference.

3.1 U-Net (placeholder)

Planned slot for the learned U-Net FG/BG mask generator. Add architecture details (depth, channels, skips), preprocessing steps, and post-processing (e.g., thresholding, hole filling) when the model is integrated.

4 Notes for Integration

All segmenters operate per 2D slice and return uint8 masks in $\{0, 1\}$. Background is always defined as the complement of foreground to ensure downstream IQMs receive matching shapes and disjoint FG/BG sets.

References

[1] Amir Reza Sadri et al. "MRQy: An open-source tool for the quality control of structural MRI images". In: *Journal of Imaging* 6.1 (2020). Placeholder entry; replace with canonical citation if available, pp. 1–15.